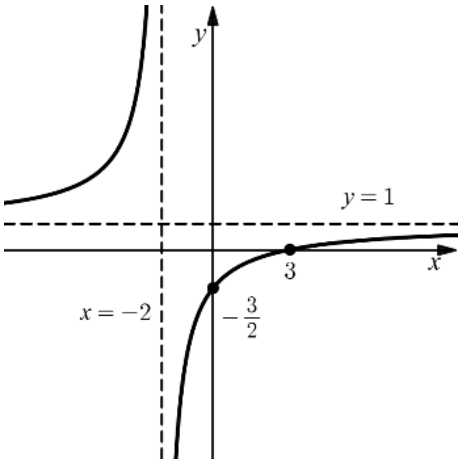
1) a)

$y = 1$ and $x = -2$

1) b)

When $x = 0$, $y = -\frac{3}{2}$

When $y = 0$, $x = 3$



1) c)

$\{x: x \in \mathbb{R}, -2 < x < 3\}$

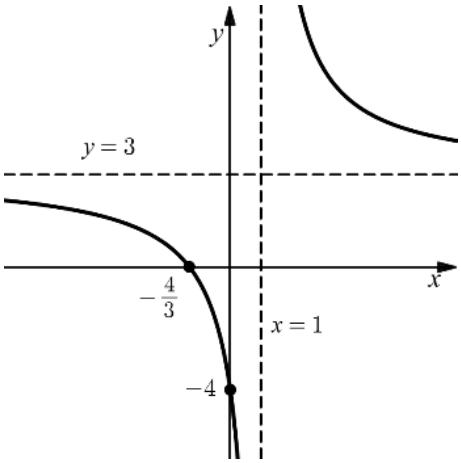
2) a)

$y = 3$ and $x = 1$

2) b)

When $x = 0$, $y = -4$

When $y = 0$, $x = -\frac{4}{3}$



2) c)

$x \in \left(\infty, -\frac{4}{3}\right] \cup (-1, \infty)$

3) a)

Asymptote of $x = 9$ gives $b = 9$

Using $(10, 15)$ then $15 = \frac{20-a}{10-9}$

$a = 5$ and $b = 9$

3) b)

When $x = 0$, $y = \frac{5}{9}$

When $y = 0$, $x = \frac{5}{2}$

$\left(\frac{5}{2}, 0\right)$ and $\left(0, \frac{5}{9}\right)$

3) c)

$\frac{2x-5}{x-9} > 0$

$\left\{x : x \in \mathbb{R}, x < \frac{5}{2}\right\} \cup \{x : x \in \mathbb{R}, x > 9\}$

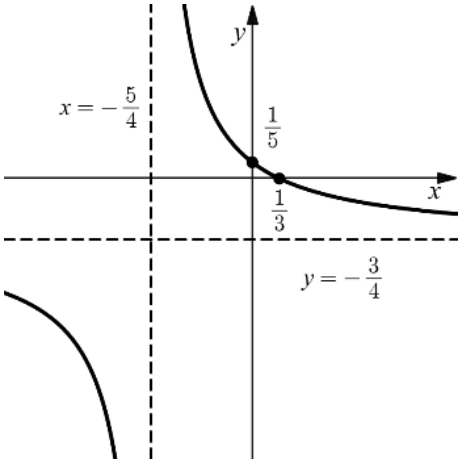
4) a)

$y = -\frac{3}{4}$ and $x = -\frac{5}{4}$

4) b)

When $x = 0$, $y = \frac{1}{5}$

When $y = 0$, $x = \frac{1}{3}$



4) c)

$\frac{1-3x}{4x+5} = \frac{1}{2} \Rightarrow 2(1-3x) = 4x+5 \Rightarrow -3 = 10x \therefore x = -\frac{3}{10}$

$\left\{x: x \in \mathbb{R}, x < -\frac{5}{4}\right\} \cup \left\{x: x \in \mathbb{R}, x > -\frac{3}{10}\right\}$

5) a)

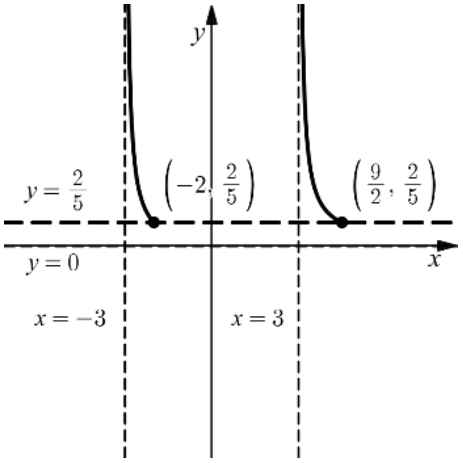
$k = 9$

5) b)

$\frac{x}{x^2-9} = \frac{2}{5} \Rightarrow 5x = 2x^2 - 18 \Rightarrow 2x^2 - 5x - 18 = 0$

$\therefore x = -2$ or $x = \frac{9}{2}$

5) c)



$\{x: x \in \mathbb{R}, -3 < x \leq -2\} \cup \left\{x: x \in \mathbb{R}, 3 < x \leq \frac{9}{2}\right\}$

6) a)

$x^2 + x - 6 = (x + 3)(x - 2)$

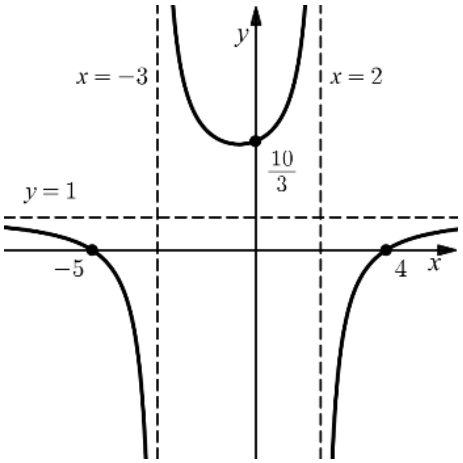
So asymptotes are

$x = -3$, $x = 2$ and $y = 1$

6) b)

When $x = 0$, $y = -\frac{10}{3}$

When $y = 0$, $x^2 + x - 20 = 0$ so $x = -5$ or $x = 4$



6) c)

$\frac{x^2+x-20}{x^2+x-6} = \frac{9}{2} \Rightarrow 2x^2 + 2x - 40 = 9x^2 + 9x - 54 \Rightarrow 7x^2 + 7x - 14 = 0$

so $x = 1$ or $x = -2$

$x \in (-\infty, 3) \cup [-2, 1] \cup (2, \infty)$

7) a)

$a = 2$

$y = \frac{2x^2+4x+b}{x^2-3x+c}$

At $(-1, 0)$

$0 = 2(-1)^2 + 4(-1) + b \therefore b = 2$

$y = \frac{2x^2+4x+2}{x^2-3x+c}$

At $\left(3, \frac{16}{3}\right)$

$\frac{16}{3} = \frac{18+12+2}{9-9+c} \therefore c = 6$

$a = 2$, $b = 2$ and $c = 6$

7) b)

$\frac{2x^2+4x+2}{x^2-3x+6} = 2 \Rightarrow 2x^2 + 4x + 2 = 2x^2 - 6x + 12 \Rightarrow 10x = 10 \therefore x = 1$

$x \in (-\infty, 1)$

8) a)

$2x^2 - 4x + 3 = 0$ has no real solutions

So the only asymptote is $y = \frac{1}{2}$

8) b)

$\frac{x^2-1}{2x^2-4x+3} = k \Rightarrow x^2 - 1 = k(2x^2 - 4x + 3)$

$\Rightarrow (1 - 2k)x^2 + 4kx - (3k + 1) = 0$

There exists solutions when:

$(4k)^2 + 4(1 - 2k)(3k + 1) \geq 0$

$16k^2 - 24k^2 + 4k + 4 \geq 0$

$2k^2 - k - 1 \leq 0$

$(2k + 1)(k - 1) \leq 0$ as req.

8) c)

When $y = -\frac{1}{2}$, $-\frac{1}{2} = \frac{x^2-1}{2x^2-4x+3} \Rightarrow 2x^2 - 4x + 3 = 2 - 2x^2$

$\Rightarrow 4x^2 - 4x + 1 = 0$ so $x = \frac{1}{2}$

When $y = 1$, $1 = \frac{x^2-1}{2x^2-4x+3} \Rightarrow 2x^2 - 4x + 3 = x^2 - 1$

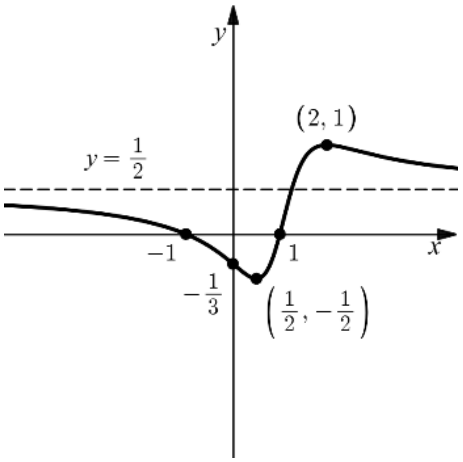
$\Rightarrow x^2 - 4x + 4 = 0$ so $x = 2$

Min is at $\left(\frac{1}{2}, -\frac{1}{2}\right)$, Max is at $(2, 1)$

8) d)

When $x = 0$, $y = -\frac{1}{3}$

When $y = 0$, $x = 1$ or $x = -1$



8) e)

$\frac{x^2-1}{2x^2-4x+3} = x - 1 \Rightarrow x^2 - 1 = (x-1)(2x^2 - 4x + 3)$

$x^2 - 1 = 2x^3 - 4x^2 + 3x - 2x^2 + 4x - 3$

$2x^3 - 7x^2 + 7x - 2 = 0$

Let $f(x) = 2x^3 - 7x^2 + 7x - 2$

$f(1) = 2 - 7 + 7 - 2 = 0$

$f(2) = 16 - 28 + 14 - 2 = 0$

$f\left(\frac{1}{2}\right) = \frac{1}{4} - \frac{7}{4} + \frac{7}{2} - 2 = 0$

$\therefore 2x^3 - 7x^2 + 7x - 2 = (x-1)(x-2)(2x-1) = 0$

$\therefore x = 1$, $x = 2$ or $x = \frac{1}{2}$

$\left\{x: x \in \mathbb{R}, x \leq \frac{1}{2}\right\} \cup \{x: x \in \mathbb{R}, 1 \leq x \leq 2\}$

9) a)

Two asymptotes which are perpendicular so of the form $y = \frac{ax+b}{cx+d}$

The asymptotes give: $y = \frac{-2x+k}{x-1}$

At $(3, 0)$ then $0 = \frac{-6+k}{3-1}$ so $k = 6$

$f(x) = \frac{-2x+6}{x-1}$

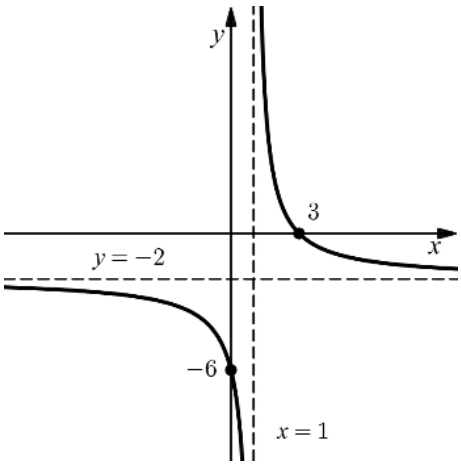
9) b)

When $x = 0$, $y = -6$

When $y = 0$, $x = 3$

$(0, -6)$ and $(3, 0)$

9) c)



9) d)

$\frac{-2x+6}{x-1} = 4 \Rightarrow -2x + 6 = 4x - 4 \Rightarrow 6x = 10$

$\therefore x = \frac{5}{3}$

$\therefore x \in \left(1, \frac{5}{3}\right]$

10) a)

The curve is of the form $y = \frac{3x^2+ax+b}{x^2+cx+2}$

Consider the points:

$-1 = \frac{b}{2}$ so $b = -2$

But also

$\frac{29}{7} = \frac{3a+25}{3c+11} \Rightarrow 319 + 87c = 21a + 175 \Rightarrow 21a - 87c = 144$

$-5 = \frac{1-a}{3-c} \Rightarrow -15 + 5c = 1 - a \Rightarrow a + 5c = 16$

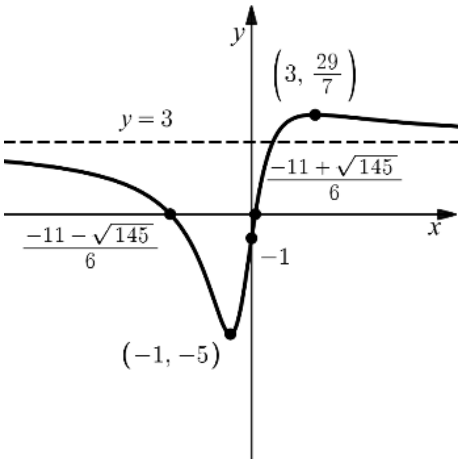
$a = 11, c = 1$

$\therefore f(x) = \frac{3x^2+11x-2}{x^2+x+2}$

10) b)

When $y = 0$, $3x^2 + 11x - 2 = 0 \therefore x = -\frac{11}{6} + \frac{\sqrt{145}}{6}$ or $x = -\frac{11}{6} - \frac{\sqrt{145}}{6}$

10) c)



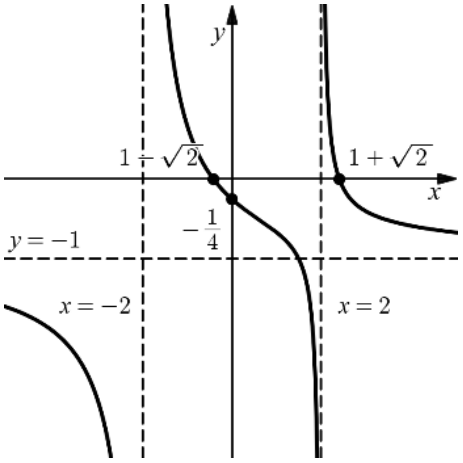
10) d)

$-1 = \frac{3x^2+11x-2}{x^2+x+2} \Rightarrow -x^2 - x - 2 = 3x^2 + 11x - 2 \Rightarrow 4x^2 + 12x = 0 \Rightarrow 4x(x + 3) = 0$

$\therefore x = 0$ or $x = -3$

$\{x: x \in \mathbb{R}, x \leq -3\} \cup \{x: x \in \mathbb{R}, x \geq 0\}$

11) a)



11) b)

Assume the function is of the form: $y = \frac{ax^2+bx+c}{px^2+qx+r}$

$\left(0, -\frac{1}{4}\right)$ gives that $\frac{c}{r} = -\frac{1}{4}$

Numerator: $a\left(x - \left(1+\sqrt{2}\right)\right)\left(x - \left(1-\sqrt{2}\right)\right) = a(x^2 - 2x - 1)$

Denominator: $p(x-2)(x+2) = p(x^2-4)$

$y = -1$ gives that $\frac{a}{p} = -1$

$\therefore a = -p$

$\frac{a(x^2-2x-1)}{-a(x^2-4)} = -\frac{x^2-2x-1}{x^2-4}$

$\therefore \mathrm{f}(x) = -\frac{x^2-2x-1}{x^2-4}$

11) c)

$-1 = -\frac{x^2-2x-1}{x^2-4} \Rightarrow x^2 - 4 = x^2 - 2x - 1 \Rightarrow 2x = 3$

$\therefore x = \frac{3}{2}$

$\therefore x \in (-\infty, -2) \cup \left[\frac{3}{2}, 2\right)$